SAFE-CARE™ Framework One-Pager

A practical operating model for patient-facing healthcare AI agents

SAFE-CARE Framework — authored by Yassen Eltayeb, Conefia · Version 1.1 · July 2026

Start with the failure: a patient says, "I have heavy bleeding and feel dizzy." The agent's first job is not to sound helpful—it is to recognize risk, avoid reassurance, and route the user to prompt care. SAFE-CARE turns that principle into an end-to-end operating model.

The SAFE-CARE operating model

Principle	Practical requirement
S — Scope	Define intended use, prohibited actions, claims, users, and escalation boundary.
A — Anchor	Ground health claims in approved evidence and make support traceable.
F — Fence	Layer input, output, retrieval, tool, memory, and monitoring guardrails.
E — Evaluate	Use simulations, ablations, AI judges, human calibration, and regression tests.
C — Contextualize	Personalize from confirmed, time-stamped, correctable health context.
A — Act	Use permissioned tools with confirmation, audit logs, and rollback.
R — Route	Escalate urgent, uncertain, severe, or regulated cases to humans.
E — Evidence	Maintain a safety case, registries, release report, and production monitoring.

Architecture: User → input safety → evidence/context → orchestrator → governed tools → output safety → response/citations → monitoring and human review.

Who benefits

Stakeholder	Value
Patients	Clear education, appropriate personalization, citations,

Stakeholder	Value
	and safe escalation.
Clinicians and operations	Lower repetitive follow-up burden while high-risk cases stay human-owned.
Product and engineering teams	A repeatable build-test-release pattern instead of one-off prompt work.
Healthcare organizations & the field	A shareable safety baseline that raises the bar for patient-facing AI.

In one line

Healthcare AI safety is not a prompt. It is an operating system: scope, evidence, guardrails, tools, memory, evaluation, escalation, and governance.

Companion documents

Published alongside this one-pager in the SAFE-CARE public release repository (github.com/Conefia/SAFE-CARE · DOI 10.5281/zenodo.21330841):

- **Menopause-support companion design case study** — the first applied case study of SAFE-CARE, based on the author's prior applied work and presented in anonymized form (docs/03_SAFE-CARE_Menopause-Support_Companion_Case_Study).
- **Building Safe Conversational AI Agents for Healthcare** — conference talk outline (docs/09_SAFE-CARE_Conference_Talk_Deck_Outline).

Public framework for educational and engineering use. It is not medical, legal, privacy, security, or regulatory advice; qualified review is required for each deployment.

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